

Efficient LLM Serving with KV Cache, Speculative Decoding, and Quantization: A Survey

Serving large language models is a systems problem shaped by memory bandwidth, accelerator utilization, latency targets, quality constraints, and rapidly changing model architectures. KV cache management, speculative decoding, and quantization have become three central levers for reducing inference cost: KV-cache techniques control memory growth during autoregressive decoding, speculative methods trade draft computation for fewer expensive target-model steps, and quantization reduces model and cache precision while trying to preserve quality. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around serving pipelines, cache hierarchy, batching and scheduling, speculative verification, low-bit inference, attention kernels, distributed deployment, compression, and evaluation. The survey argues that efficient LLM serving requires co-design across algorithms, kernels, memory managers, schedulers, and quality monitors.

Keywords:

LLM serving; KV cache; speculative decoding; quantization; batching; attention kernels; distributed inference; latency

1 Introduction

Large language model (LLM) serving has become one of the dominant costs in modern AI systems. Training produces a model, but serving determines whether that model can respond to millions of heterogeneous requests with acceptable latency, quality, and cost. The bottlenecks are not only algorithmic. They involve GPU memory, memory bandwidth, scheduling, network communication, batching, kernel fusion, precision, and workload burstiness [51, 68, 101, 127, 140, 182, 243, 249].

KV cache management, speculative decoding, and quantization are especially important because they attack different parts of the decoding pipeline. KV cache techniques reduce the memory footprint of autoregressive context. Speculative decoding reduces the number of expensive target-model steps by verifying tokens proposed by a cheaper draft process. Quantization reduces parameter, activation, and sometimes cache precision. Each technique

changes the serving stack, and the best system usually combines all three with batching, scheduling, and optimized attention kernels [51, 68, 106, 127, 140, 182, 243, 249].

This survey reviews efficient LLM serving through the lens of systems co-design. We focus on how algorithmic speedups interact with memory managers, GPU kernels, distributed execution, and quality constraints.

2 Background

2.1 The LLM Serving Pipeline

An LLM request usually has a prefill phase, where the prompt is processed in parallel, and a decode phase, where tokens are generated autoregressively. Prefill is compute intensive; decode is often memory-bandwidth bound because each step loads weights and cache state for a small amount of new computation. Serving systems therefore optimize for both

throughput and tail latency, with different trade-offs for chat, batch generation, agents, retrieval-augmented workloads, and long-context applications [51, 68, 101, 127, 140, 182, 243, 249].

2.2 KV Cache

The key-value cache stores attention keys and values from previous tokens so that decoding does not recompute the entire context at every step. The cache grows with sequence length, batch size, number of layers, and hidden dimension. It can become the dominant memory consumer during long-context serving. Cache paging, prefix reuse, eviction, compression, and quantization are therefore central to practical deployment [51, 68, 106, 127, 140, 182, 243, 249].

2.3 Speculative Decoding

Speculative decoding uses a draft generator to propose multiple tokens, then verifies them with the target model. When draft tokens are accepted, the target model advances several positions per expensive call. The speedup depends on draft quality, verification cost, acceptance rate, batch interaction, and memory traffic. Recent variants use small models, heads attached to the target model, multi-token prediction, tree drafts, or retrieval-assisted proposals [83, 101, 104, 117, 161, 180, 208, 275].

2.4 Quantization

Quantization reduces numerical precision for weights, activations, or caches. Weight-only quantization primarily reduces model memory and bandwidth; activation and KV-cache quantization further reduce runtime state; FP8 and INT4 methods trade hardware support, calibration, and accuracy. Serving needs quantization methods that preserve quality under long contexts, sampling, batching, and mixed workloads [68, 95, 117, 122, 182, 208, 243, 266].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with LLM serving, inference serving, KV cache, key-value cache, PagedAttention, prefix caching, continuous batching, request

scheduling, speculative decoding, speculative sampling, draft models, multi-token prediction, Medusa, EAGLE, lookahead decoding, LLM quantization, GPTQ, AWQ, SmoothQuant, weight-only quantization, activation quantization, FP8, INT4, FlashAttention, attention kernels, prefill, decode kernels, GPU inference, distributed inference, tensor parallel execution, offloading, compression, pruning, sparsity, and energy-efficient inference. Records were retained when title or abstract evidence connected LLMs or large language models with inference efficiency, serving systems, cache management, decoding acceleration, quantization, kernels, scheduling, distributed deployment, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize efficient LLM serving along six axes. The first is *phase*: prefill, decode, or end-to-end request execution. The second is *resource bottleneck*: memory capacity, memory bandwidth, compute, network, or scheduler overhead. The third is *algorithmic lever*: KV-cache management, speculative verification, quantization, compression, pruning, or attention approximation. The fourth is *system lever*: batching, admission control, parallelism, offloading, kernel fusion, or cache placement. The fifth is *quality constraint*: perplexity, task accuracy, sampling distribution, safety behavior, or long-context fidelity. The sixth is *workload*: interactive chat, batch inference, agents, retrieval-augmented generation, code, or multimodal serving.

3.3 KV Cache Management

KV-cache systems manage a tension between reuse and memory pressure. Prefix caching can reuse shared prompts, paging can reduce fragmentation, and eviction or compression can extend context length. Cache quantization reduces memory traffic but can alter attention scores. Future cache managers will likely become workload-aware: they will distinguish reusable system prompts, retrieval prefixes, conversational history, and low-value stale context [51, 68, 106, 127, 140, 182, 243, 249].

3.4 Speculative Decoding Families

Speculative methods differ in the source of draft tokens. A small draft model can be fast but may have low acceptance; target-attached heads reduce serving complexity; tree proposals increase parallelism; retrieval or n-gram drafts exploit repetition. The

systems question is whether draft computation, verification, cache updates, and batching actually reduce end-to-end latency under realistic workloads [83, 101, 104, 117, 161, 180, 208, 275].

3.5 Quantization for Serving

Quantization methods must be judged by deployed behavior, not only benchmark perplexity. Calibration data, outlier handling, group size, activation scaling, and hardware support all matter. Weight-only methods often improve memory bandwidth, while activation and cache quantization reduce runtime state. The hardest cases are long contexts, reasoning models, multilingual text, code, and workloads where small quality changes are unacceptable [68, 95, 117, 122, 182, 208, 243, 266].

3.6 Batching and Scheduling

Serving systems use batching to increase accelerator utilization, but batching interacts with variable sequence lengths, decode steps, cache memory, fairness, and tail latency. Continuous batching admits and retires requests dynamically. Schedulers must decide which requests share a batch, how much memory to reserve, when to preempt, and how to trade throughput against latency objectives [51, 65, 95, 101, 106, 122, 140, 148].

3.7 Attention Kernels and Decode Optimization

Optimized kernels reduce overhead in attention, prefill, decode, normalization, and matrix multiplication. FlashAttention and related methods improve memory locality for attention, while decode kernels optimize small-batch autoregressive steps. Kernel design is inseparable from cache layout, quantization format, tensor parallelism, and accelerator hardware [51, 68, 101, 106, 127, 182, 243, 249].

3.8 Distributed Inference and Offloading

Large models often require tensor parallelism, pipeline parallelism, expert parallelism, or offloading to CPU, disk, or remote memory. Distributed inference adds communication overhead and scheduling complexity. Offloading can make larger models feasible on limited hardware but can also hurt latency if transfers are not hidden behind computation [68, 95, 101, 140, 148, 210, 243, 266].

3.9 Compression, Pruning, and Efficient Adaptation

Quantization is only one form of compression. Pruning, sparsity, distillation, low-rank approximation, and mixture-of-experts routing can also reduce serving cost. These methods interact with serving differently: sparsity requires hardware and kernel support, distillation changes model quality, and mixture-of-experts models shift the bottleneck toward routing and communication [51, 68, 101, 127, 140, 182, 243, 249].

3.10 Evaluation

Efficient serving should be evaluated with end-to-end metrics: time to first token, inter-token latency, throughput, memory footprint, energy, cost, quality degradation, and tail latency. A method that improves synthetic decode speed may fail under mixed prompt lengths, long contexts, streaming outputs, or bursty traffic. Reproducible benchmarks should therefore specify hardware, model, workload, batch policy, context distribution, and quality tests [51, 68, 101, 106, 140, 210, 243, 249].

3.11 Representative Literature Map

The following map cites the full retained corpus by theme, making the coverage explicit across cache, decoding, quantization, scheduling, kernels, distribution, compression, and evaluation.

Kv cache management and memory hierarchy The kv cache management and memory hierarchy cluster contains work that develops methods, systems, datasets, or analyses central to this survey [91, 95, 127, 133, 164, 165, 176, 204, 276]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [23, 29, 42, 113, 115, 142, 144, 207, 254]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [20, 138, 139, 160, 171, 190, 213, 214, 274]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [84, 116, 151, 173, 174, 196, 209, 221, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [35, 66, 183, 238, 242, 251, 265, 282, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 40, 41, 81, 102, 103,

145, 223, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [93, 110, 112, 156, 212, 250, 264, 268, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 32, 59, 64, 78, 131, 150, 222, 234]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [58, 125, 257, 262, 296].

Speculative decoding and draft verification The speculative decoding and draft verification cluster contains work that develops methods, systems, datasets, or analyses central to this survey [98, 101, 143, 184, 192, 208, 227, 232, 275]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [45, 72, 129, 233, 269].

Quantization and low-bit inference The quantization and low-bit inference cluster contains work that develops methods, systems, datasets, or analyses central to this survey [69, 130, 181, 182, 201, 205, 219].

Batching, scheduling, and serving systems The batching, scheduling, and serving systems cluster contains work that develops methods, systems, datasets, or analyses central to this survey [51, 65, 99, 104, 106, 109, 117, 180, 256]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [39, 61, 92, 111, 120, 152, 161, 278, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 79, 118, 149, 167, 218, 229, 259, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [48, 49, 89, 100, 123, 261, 281, 287, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 11, 12, 18, 124, 168, 178, 200, 220]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 67, 71, 97, 132, 136, 230, 236, 246]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 28, 75, 146, 157, 172, 186, 194, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions

[2, 50, 63, 211, 226, 231, 260, 286, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 56, 88, 134, 135, 141, 188, 240, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 33, 34, 70, 107, 153, 197, 198, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 169, 185, 187, 225].

Attention kernels and decoding optimization The attention kernels and decoding optimization cluster contains work that develops methods, systems, datasets, or analyses central to this survey [37, 68, 114, 122, 202, 210, 249, 267, 297]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 5, 9, 36, 47, 76, 137, 177, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 27, 52, 53, 74, 87, 175, 216, 272]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 22, 26].

Distributed inference and offloading The distributed inference and offloading cluster contains work that develops methods, systems, datasets, or analyses central to this survey [62, 83, 148, 162, 163, 243, 247, 290, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [13, 82, 108, 155, 189, 237, 258, 295].

Compression, pruning, and efficient adaptation The compression, pruning, and efficient adaptation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [31, 80, 86, 90, 154, 170, 195, 253, 280]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [19, 38, 191, 217, 241, 271, 273, 277].

Benchmarks, quality, and energy evaluation The benchmarks, quality, and energy evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [44, 73, 121, 126, 140, 158, 166, 228, 266]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [14, 43, 77, 85, 119, 203, 224, 245, 300]. Additional retained studies broaden the cluster with alternative

architectures, training signals, data sources, and deployment assumptions [17, 60, 128, 147, 193, 206, 248, 252, 255]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [16, 94, 96, 105, 159, 215, 235, 239, 244]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 25, 54, 55, 179, 199, 263].

4 Challenges and Prospects

4.1 Co-Design Across the Serving Stack

KV cache compression, speculative decoding, quantization, and scheduling cannot be optimized independently. A draft method changes cache updates; quantization changes kernel choices; batching changes speculative acceptance dynamics. Future systems need integrated optimization rather than isolated speedup claims.

4.2 Long Context and Memory Pressure

Long-context models make cache memory a first-class resource. Serving systems will need adaptive cache placement, loss-aware compression, prefix reuse, and policies that decide which context should remain exact, approximate, or evicted.

4.3 Quality Under Acceleration

Serving optimizations are useful only when quality remains acceptable. Quantization, approximate attention, and speculative verification should be tested on reasoning, code, safety, multilingual, and long-context workloads rather than only on short perplexity benchmarks.

4.4 Workload-Aware Scheduling

Real workloads include interactive users, batch jobs, agents, retrieval pipelines, and multimodal requests. Schedulers should adapt to request deadlines, cache reuse opportunities, output length uncertainty, and fairness constraints.

4.5 Toward Self-Optimizing Serving Systems

Future serving systems may monitor workloads, profile kernels, choose quantization modes, allocate

cache, and tune speculative draft strategies online. Such systems will need guardrails so that automatic optimization does not silently degrade quality or safety.

5 Conclusion

Efficient LLM serving is a co-design problem across algorithms and systems. KV cache techniques manage memory, speculative decoding reduces target-model steps, quantization reduces precision and bandwidth, batching improves utilization, kernels exploit hardware locality, and distributed execution expands model scale. This survey reviewed these components and their interactions. The central lesson is that end-to-end serving performance depends on the whole stack: memory layout, scheduler policy, numerical format, draft quality, kernel design, and workload shape all matter together.

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